MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
49RH PROCESS PLANT:HELIPAD
DESIGN OF HELIPAD TLFO GRADE SLAB

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute
TLOF	:	Touchdown & Liftoff Area
FATO	:	Final Approach & Takeoff Area

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Helipad Structure.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Helipad Structure only.

4. DESIGN METHODOLOGY & ASSUMPTIONS

A helipad is a landing area or platform for helicopters and powered lift aircraft. While helicopters and powered lift aircraft are able to operate on a variety of relatively flat surfaces, a helipad provides a clearly marked hard surface away from obstacles where such aircraft can land safely. The Helipad in this area has been designed for the specifications of **AW139** Vehicle.

TLOF, FATO, lighting, Mooring, Fire Prevention, Safety & all the applicable specifications have been followed as per Chapter-2, 3, 4, 5, 6 & 9 respectively from Aviation Code & Standards.

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4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] MOI Directorate of Safety - SPECIFICATION GUIDE FOR HELICOPTERS

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

4.3.1. Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for foundations.

4.3.2. Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

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5. DESIGN LOADS

5.1. Dead Load



OVERVIEW

The AgustaWestland AW139 is one of the most modern and efficient helicopters in the world. The AW139 provides unparalleled mission flexibility and passenger comfort. This state-of-the-art aircraft is equipped with the latest safety features and sets the standard for its class.

PERFORMANCE

The AW139 is engineered to the highest safety standards with energy-absorbing landing gear, fuselage and seats designed to meet the rigorous JAR/FAR 29 standards. The cockpit features excellent visibility, redundant flight systems and Honeywell digital avionics, reducing pilot workload. The AW139 delivers Category A (Class 1) performance from the helipad at maximum gross weight, allowing continued flight in the event of an engine failure.

Cruise Airspeed	150 kt
Basic Operating Weight	10,100 lb
Max Gross Weight	14,992 lb
Max Fuel	2,765 lb (950 lb hr / 3.0 hrs)
Max Fuel w Aux	3,647 lb (950 lb hr / 3.8 hrs)
Fuel Burn	150 gallons/hr

STANDARD CONFIGURATION

12 Passenger
Air Conditioning

SAFETY FEATURES

Traffic Collision Avoidance System
GPS Position Tracking (Sky Connect MMU-II)
High-Visibility Blades
Pulse Lights/LED Strobe Lights
406 Emergency Location Transmitter
Emergency Floatation (water activation) with External Rafts
Underwater Locator Beacons
CVR/FDR-FDM
4-Axis Autopilot
EGPWS
Cat A Certified
Dual GPS
Radar Altimeter (Aural Warning)
Weather Radar
WAAS
ADSB Out
HUMS

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Consider minimum Impact factor of 1.50 as per Clause C.4.11.2 of ASCE/SEI-7 attached below.

Although these structures are intended to be specifically kept clear of nonhelicopter occupant loads on the landing and taxi areas, the uniform load requirement is a minimum to ensure a degree of substantial construction and the potential to resist the effects of unusual events.

Additional information on helipad design can be found in Annex 14 to the Convention on International Civil Aviation, Aerodromes, Volume II (ICAO 2013).

C4.11.2 Concentrated Helicopter Loads. Concentrated loads applied separately from the distributed loads are intended to cover the primary helicopter loads. The designer should always consider the geometry of the design basis helicopter for applying the design loads. A factor of 1.5 is used to address impact loads (two single concentrated loads of 0.75 times the maximum takeoff weight) to account for a hard landing with many kinds of landing gear. The designer should be aware that some helicopter configurations, particularly those with rigid landing gear, could result in substantially higher impact factors that should be considered.

The 3,000-lb (13.35-kN) concentrated load is intended to cover maintenance activities, similar to the jack load for a parking garage.

C4.13 LIBRARY STACK ROOMS

Where library shelving installation does not fall within the parameter limits that are specified in Section 4.13 and Table 4.3-1, the design should account for the actual conditions. For example, the floor loading for storage of medical X-ray film may easily exceed 200 psf (9.58 kN/m^2), mainly because of the increased depth of the shelves. Mobile library shelving that rolls on rails should also be designed to meet the actual requirements of the specific installation, which may easily exceed 300 psf (14.4 kN/m^2). The rail support locations and deflection limits should be considered in the design, and the engineer should work closely with the system manufacturer to provide a serviceable structure.

C4.14 SEATING FOR ASSEMBLY USES

The lateral loads apply to "stadiums and arenas" and to "reviewing stands, grandstands, and bleachers." However, it does not apply to "gymnasiums—main floors and balconies." Consideration should be given to treating gymnasium balconies that have stepped floors for seating as arenas, and requiring the appropriate swaying forces.

C4.17 SOLAR PANEL LOADS

C4.17.1 Roof Loads at Solar Panels. These provisions are added to the 2016 edition of the standard to address the installation of rooftop solar panels consistent with current practices (Blaney and LaPlante 2013). These provisions allow the offset of roof live load where the space below the solar panel is considered inaccessible. The dimension of 24 in. (610 mm) was chosen as the clear vertical distance as it is consistent with existing published requirements for solar panel systems and is also a typical minimum height permitted for access into or out of spaces.

C4.17.3 Open-Grid Roof Structures Supporting Solar Panels. This section reduces the uniform roof live load for building structures such as carports and shade structures, which do not include roof deck or sheathing, to the value of the minimum uniform roof live load permitted by Section 4.8.2. The concentrated roof live load requirement in Table 4.3-1 is not modified by this section.

REFERENCES

American Association of State Highway and Transportation Officials (AASHTO). *LRFD bridge design specifications*, 7th Ed., 2014, with 2015 interim revisions, AASHTO, Washington, DC.

American Society of Mechanical Engineers (ASME). (2013). *American national standard safety code for elevators and escalators*. ASME, New York, A17.

Association of Iron and Steel Technology (AIST). (2003). "Guide for the design and construction of mill buildings." *Tech. Report No. 13*, AISE, Warrendale, PA.

Blaney, C., and LaPlante, R. (2013). "Recommended design live loads for rooftop solar arrays." *Proc., SEAOC Convention*, 264–278.

Chalk, P. L., and Corotis, R. B. (1980). "Probability model for design live loads." *J. Struct. Div.*, 106(10), 2017–2033.

Corotis, R. B., Harris, J. C., and Fox, R. R. (1981). "Delphi methods: Theory and design load application." *J. Struct. Div.*, 107(6), 1095–1105.

Ellingwood, B. R., and Culver, C. G. (1977). "Analysis of live loads in office buildings." *J. Struct. Div.*, 103(8), 1551–1560.

Harris, M. E., Bova, C. J., and Corotis, R. B. (1981). "Area-dependent processes for structural live loads." *J. Struct. Div.*, 107(5), 857–872.

International Civil Aviation Organization (ICAO). (2013). *Annex 14 to the convention on international civil aviation, aerodromes, Vol. II: Heliports*.

ICC/ANSI. (2009). "Accessible and Usable Buildings and Facilities." *A117.1*. International Code Council, Washington, D.C.

McGuire, R. K., and Cornell, C. A. (1974). "Live load effects in office buildings." *J. Struct. Div.*, 100(7), 1351–1366.

Metal Building Manufacturers Association (MBMA). (2012). *Metal building systems manual*, MBMA, Cleveland, OH.

MHI. (2010a). "Specifications for top running bridge and gantry type multiple girder electric overhead traveling cranes." *No. 70-2010*, MHI, Charlotte, NC.

MHI (2010b). "Specifications for top running and under running single girder electric overhead traveling cranes utilizing under running trolley hoist." *No. 74-2010*. MHI, Charlotte, NC.

MHI. (2009). *"Specifications for patented track underhung cranes and monorail systems," No. MH 27.1-2009*, MHI, Charlotte, NC.

Occupational Safety and Health Administration (OSHA). (2014a). "Code of federal regulations, Section 1910.27." *OSHA Standards*, Washington, DC.

OSHA. (2014b). "Powered platforms for building maintenance." *Code of federal regulations, Section 1910.66*. OSHA Standards, Washington, DC.

OSHA. (2014c). "Safety standards for scaffolds used in the construction industry." *Code of federal regulations, Section 1926.451*. OSHA Standards, Washington, DC.

Peir, J. C., and Cornell, C. A. (1973). "Spatial and temporal variability of live loads." *J. Struct. Div.*, 99(5), 903–922.

Sentler, L. (1975). "A stochastic model for live loads on floors in buildings." *Report No. 60*, Lund Institute of Technology, Division of Building Technology, Lund, Sweden.

Wen, Y. K., and Yeo, G. L. (2001). "Design live loads for passenger cars parking garages." *J. Struct. Eng.*, 127(3), 280–289.

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Maximum Operating Gross Weight(Including Passengers) of AW139 is **14998 Lbs**.(Refer Page-6 for Weight Details).

Therefore Total Load(Including Impact Factor as per Clause-C4.11.2 in Page-7) = **22500 lbs. (or) 100 Kns** approx.

Point Load on Each Wheel Base = 100/3 = **35 Kns** approx.

5.2. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

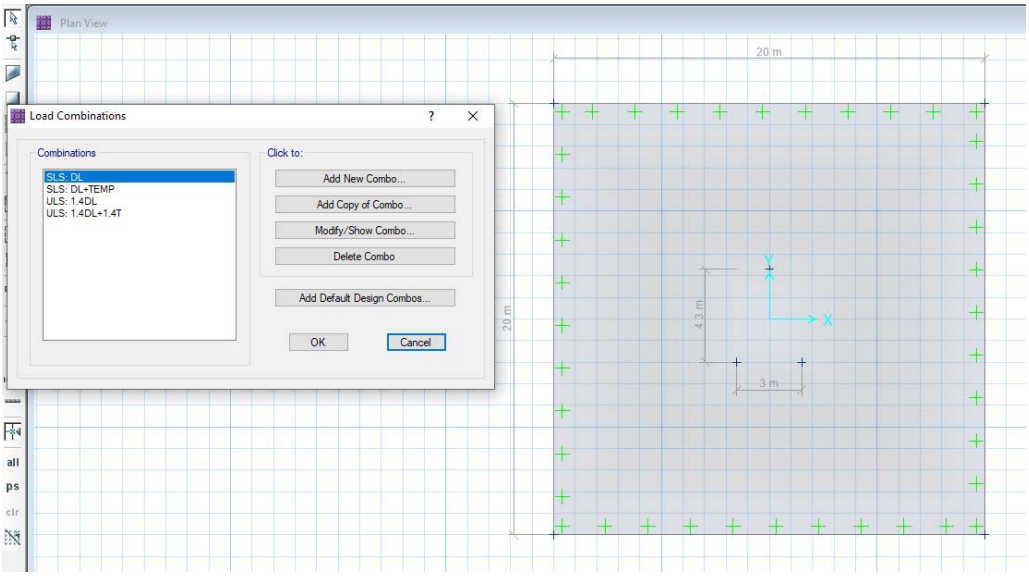
1.4D + 1.4T

Where:

D = Gross Operating Weight of Helicopter(Including Impact Factor)

T = Temperature Loads

The Above Combinations have been generated in the Foundation Design Software accordingly as shown below.



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7. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations for applicable SBC of 570 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008 for Details)

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

REFERENCES

[1]HELIPAD LAYOUT & DETAIL — MD-216-49RH-EG-CV-GAD-1006
[2]HELIPAD DETAILS OF FLOOR — MD-216-49RH-EG-CV-GAD-1007
[3]GEOTECHNICAL REPORT — MD-216-3000-EG-CV-RPT-1008
[4]STRUCTURAL DESIGN CRITERIA — MD-216-0000-EG-ST-CRT-0001

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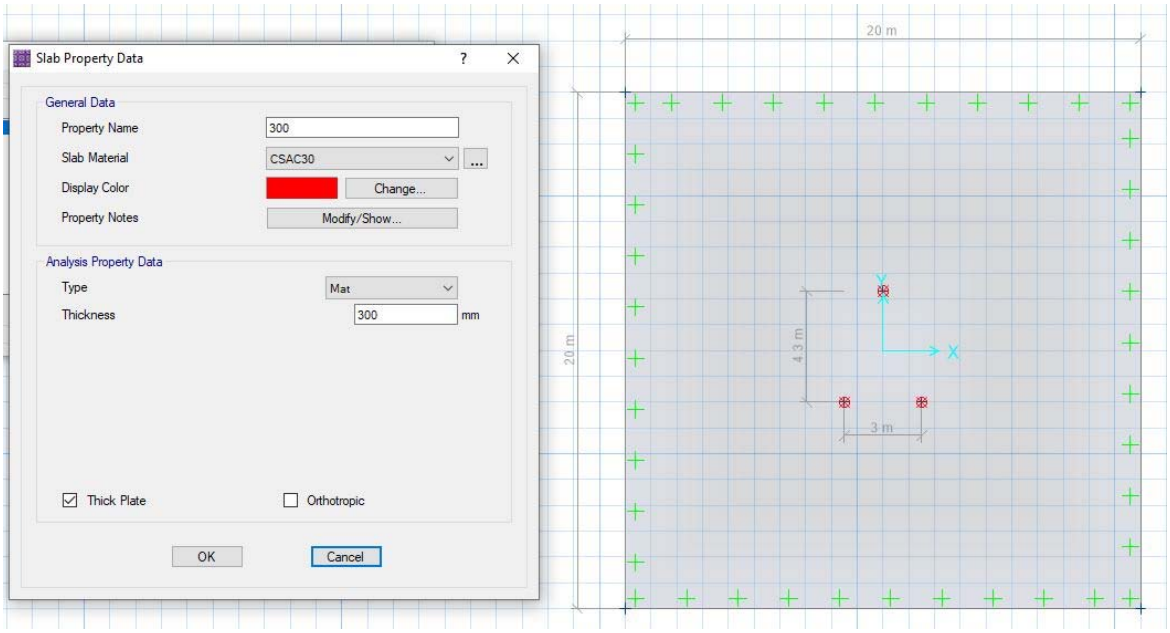
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ATTACHMENT 1 FOUNDATION DESIGN

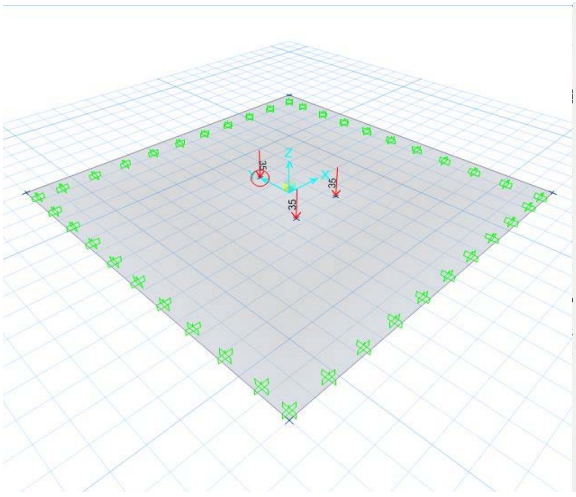
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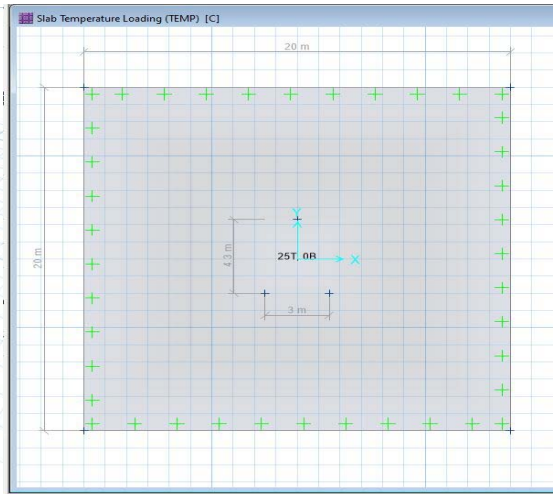
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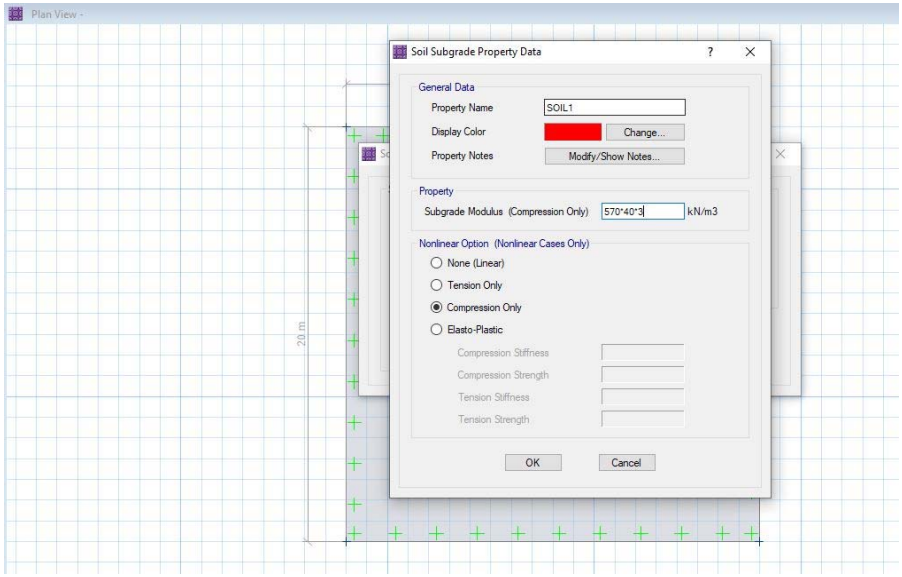
BASE DIMENSIONS FOR **300mm** THICK HELIPAD



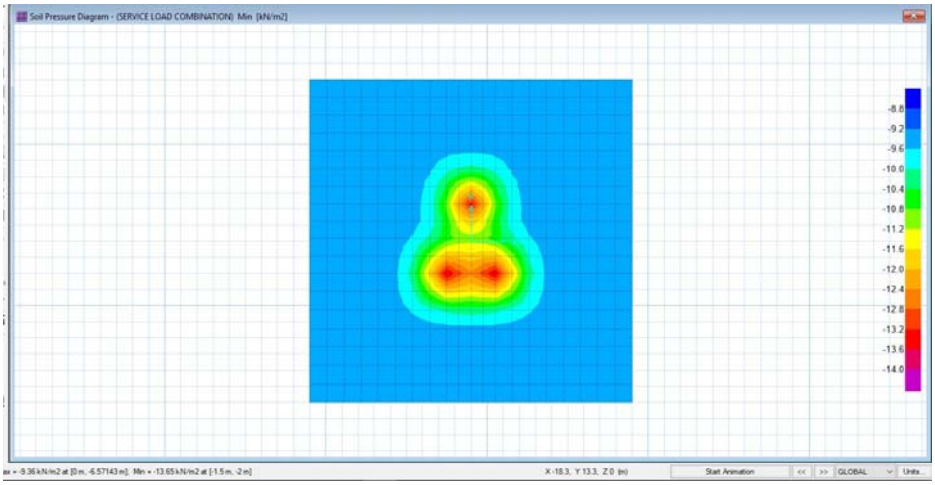
HELICOPTER WHEEL LOAD APPLIED
(35 Kn on Each Wheel)



DIFFERENTIALTEMPERATURE
LOAD APPLIED-25 Degree. Celsius



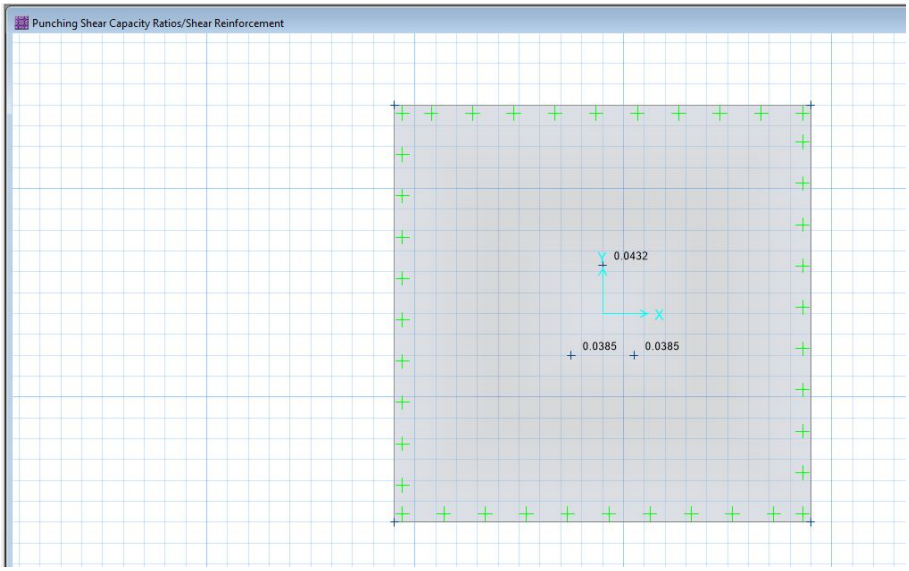
SOIL SUBGRADE MODULUS(K) APPLIED BELOW HELIPAD.



SOIL PRESSURE DIAGRAM

The Maximum Base pressure attained is 15 Kpa<<<<<150 Kpa.

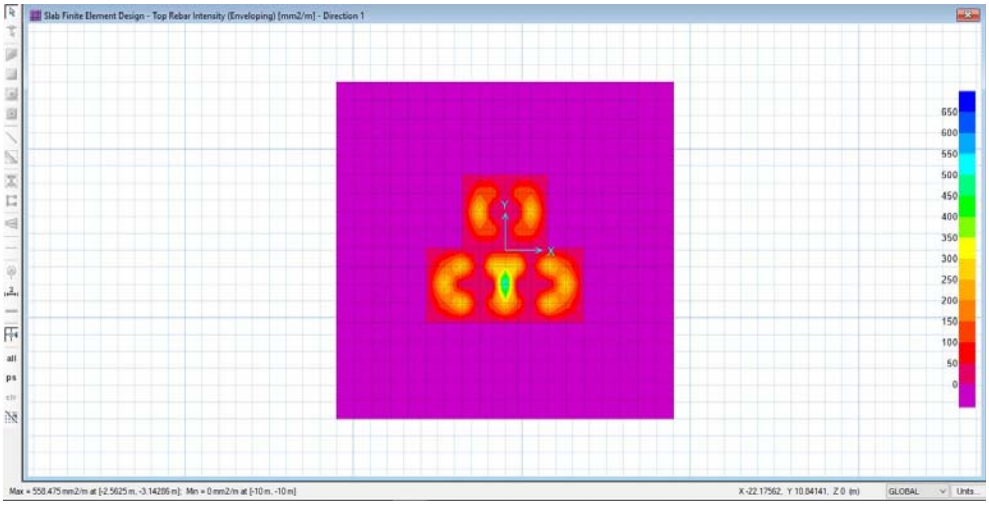
Therefore the Helipad is safe considering Base pressure check.



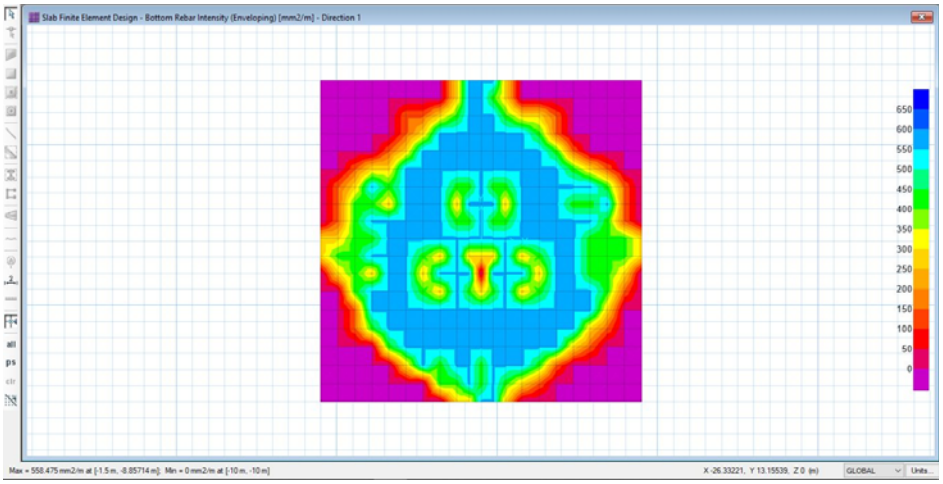
PUNCHING SHEAR RATIO CHECK

The Punching Shear ratio attained is less than 1

Therefore the Helipad is safe considering Punching shear check.



REQUIRED TOP REINFORCEMENT.



REQUIRED BOTTOM REINFORCEMENT.

So required Reinforcement T&B is 556 mm2/m. Provided Reinforcement is Dia16@200 T&B i.e., 1000 mm2/m.

Therefore the Helipad is safe considering reinforcement check criteria.

ATTACHMENT 2 AW-139 HELICOPTER SPECIFICATIONS

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OVERVIEW

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STANDARD CONFIGURATION

12 Passenger
Air Conditioning

SAFETY FEATURES

Traffic Collision Avoidance System
GPS Position Tracking (Sky Connect MMU-II)
High-Visibility Blades
Pulse Lights/LED Strobe Lights
406 Emergency Location Transmitter
Emergency Floatation (water activation) with External Rafts
Underwater Locator Beacons
CVR/FDR-FDM
4-Axis Autopilot
EGPWS
Cat A Certified
Dual GPS
Radar Altimeter (Aural Warning)
Weather Radar
WAAS
ADSB Out
HUMS

Safe, Efficient, Reliable

